



Year 11 Mathematics Extension 1

Task 2, June 2012

Time allowed : 60 minutes

Instructions:

- Full working should be shown in every question.
- Use black or blue pen to write your solution
- No liquid paper/tape to be used to correct your solution.

Question No	Question	Marks
1	If $\tan \alpha = \frac{4}{3}$ and $\cos \beta = \frac{12}{13}$, where $0 < \beta < \alpha < 90^\circ$ find the exact value of $\cos(\alpha - \beta)$.	3
2	Solve for θ where $0^\circ \leq \theta \leq 360^\circ$ a) $\tan \theta = \sin 2\theta$ b) $2 + \cos \theta = \cos 2\theta$	3 3
3	a) Express $\sin x - \cos x$ in the form of $A \sin(x - y)$. b) Hence solve $\sin x - \cos x = 1$ for x where $0^\circ \leq x \leq 360^\circ$	2 3
4	a) Find the acute angle between the lines $x - 2y = 6$ and $y = 3x - 1$. Give your answer to the nearest minute. b) Find the values of k if the line $2x - y + 1 = 0$ and $kx - 7y + 5 = 0$ make an angle 45° at their intersection.	2 2
5	a) AB is an interval joining points A (1,2) and B(7,8). Point P is on AB such that $AP = 2PB$. Find the coordinates of the point P. b) Find the coordinates of the point which divides the interval joining (3,-2) and (-5,4) externally in the ratio 5:3.	2 2
6	a) Show that $\tan(45^\circ + x) = \frac{\cos x + \sin x}{\cos x - \sin x}$ b) Hence or otherwise give the smallest positive solution of $\frac{\cos 2x + \sin 2x}{\cos 2x - \sin 2x} = \sqrt{3}$	2 2
7	a) If $\tan \frac{\theta}{2} = t$, express $\frac{1 + \sin \theta + \cos \theta}{1 + \sin \theta - \cos \theta}$ in terms of t . b) Using t-method solve $5 \cos \theta + 6 \sin \theta = -5$ for $0^\circ \leq \theta \leq 360^\circ$	3 3
8	Given $\sin 18^\circ = \frac{\sqrt{5}-1}{4}$, find $\cos 36^\circ$ in exact form.	2
9	a) Expand $(x + y)(x^2 - xy + y^2)$ b) Prove that $\cos^6 \beta + \sin^6 \beta = \frac{1}{4} + \frac{3}{4} \cos^2 2\beta$	1 3
10	a) Show that $\sin 3\alpha = 3 \sin \alpha - 4 \sin^3 \alpha$ b) Hence show that $12 \sin^3 \alpha + 2 \cos 3\alpha - 9 \sin \alpha + \sin 3\alpha = 2\sqrt{2} \cos(3\alpha + 45^\circ)$ c) Hence solve $12 \sin^3 \alpha + 2 \cos 3\alpha - 9 \sin \alpha + \sin 3\alpha = 2$ for $0^\circ \leq \alpha \leq 360^\circ$	2 3 2
End of Task 2		

① $\tan \alpha = \frac{4}{3}$ $\cos \beta = \frac{12}{13}$

$$\begin{aligned}\cos(\alpha - \beta) &= \cos \alpha \cos \beta + \sin \alpha \sin \beta \\ &= \frac{3}{5} \cdot \frac{12}{13} + \frac{4}{5} \cdot \frac{5}{13} \\ &= \frac{56}{65} \checkmark\end{aligned}$$

② a) $\tan \theta = \sin 2\theta$

$$\frac{\sin \theta}{\cos \theta} = 2 \sin \theta \cos \theta$$

$$\sin \theta (1 - 2 \cos^2 \theta) = 0 \checkmark$$

$$\begin{aligned}\swarrow \quad \searrow \\ \sin \theta = 0 & \quad \cos^2 \theta = \frac{1}{2} \\ 0^\circ, 180^\circ, 360^\circ & \quad \cos \theta = \pm \frac{1}{\sqrt{2}} \\ & \quad 45^\circ, 135^\circ, 225^\circ, 315^\circ \checkmark\end{aligned}$$

b) $2 + \cos \theta = \cos 2\theta$
 $2 \cos^2 \theta - \cos \theta - 3 = 0$
 $(2 \cos \theta - 3)(\cos \theta + 1) = 0 \checkmark$

$$\begin{aligned}\swarrow \quad \searrow \\ \cos \theta = \frac{3}{2} & \quad \cos \theta = -1 \\ \text{No solution} & \quad 180^\circ \checkmark\end{aligned}$$

③ a) $\sin x - \cos x$

$$= \sqrt{2} \left(\frac{1}{\sqrt{2}} \sin x - \frac{1}{\sqrt{2}} \cos x \right)$$

$$= \sqrt{2} (\sin x \cos y - \cos x \sin y)$$

$$= \sqrt{2} \sin(x - y) \quad \begin{matrix} \text{where } \tan y = \frac{1}{1} \checkmark \\ \text{where } \tan y = 1 \end{matrix}$$

b) $\sqrt{2} \sin(x - y) = 1 \checkmark$

$$\sin(x - y) = \frac{1}{\sqrt{2}}$$

acute $x - y = 45^\circ \checkmark$

In $\triangle ABC$ $\Rightarrow y = 45^\circ$

$x - y = 45^\circ, 315^\circ$

$x = 90^\circ, 180^\circ \checkmark$ where $y = 45^\circ$

4) a) $2y = x - 6$ — (1)
 $y = \frac{1}{2}x - 3$
 $y = 3x - 1$ — (2)

$$\tan \alpha = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \text{ where}$$

m_1, m_2 are the gradients of the lines

$$= \left| \frac{3 - \frac{1}{2}}{1 + 3 \cdot \frac{1}{2}} \right| = \frac{\frac{5}{2}}{\frac{5}{2}} = 1$$

$$\therefore \alpha = 45^\circ \text{ or } 135^\circ$$

b) $y = 2x + 1$; $y = \frac{k}{7}x + \frac{5}{7}$
 $m_1 = 2$ $m_2 = \frac{k}{7}$

$$\therefore 1 = \left| \frac{2 - \frac{k}{7}}{1 + 2 \cdot \frac{k}{7}} \right| \checkmark$$

$$1 = \left| \frac{14 - k}{7 + 2k} \right|$$

$$\frac{14 - k}{7 + 2k} = 1 \Rightarrow 3k = 7 \therefore k = \frac{7}{3}$$

$$-\frac{(14 - k)}{7 + 2k} = 1 \Rightarrow 14 - k = -7 - 2k$$

$$k = -21$$

Possible values for $k = \frac{7}{3}, -21 \checkmark$

5) A $\xrightarrow{2}$ $\xrightarrow{1}$ B
 $(1, 2) \xrightarrow{\quad} P \xrightarrow{\quad} (7, 8)$

$$P \left(\frac{2 \times 7 + 1 \times 1}{\sqrt{3}}, \frac{2 \times 8 + 1 \times 2}{\sqrt{3}} \right)$$

$$P(5, 6)$$

b) A(3, -2) B(-5, 4)

$$\left(\frac{5 \times -5 + (-3) \times 3}{5 - 3}, \frac{5 \times 4 + (-3) \times -2}{5 - 3} \right)$$

$$\left(-\frac{25-9}{2}, \frac{26}{2}\right)$$

$$= (-\sqrt{17}, \sqrt{13})$$

$$\begin{aligned} \textcircled{6} \text{ a) } \tan(45^\circ + x) &= \frac{\tan 45^\circ + \tan x}{1 - \tan 45^\circ \tan x} \\ &= \frac{1 + \tan x}{1 - \tan x} \checkmark \\ &= \frac{1 + \frac{\sin x}{\cos x}}{1 - \frac{\sin x}{\cos x}} \checkmark \\ &= \frac{\cos x + \sin x}{\cos x - \sin x} \\ &= \text{RHS.} \end{aligned}$$

$$\begin{aligned} \text{b) } \tan(45^\circ + 2x) &= \sqrt{3} \checkmark \\ 45^\circ + 2x &= \tan^{-1}(\sqrt{3}) \\ \text{acute } \theta &= 60^\circ \\ 2x &= 15^\circ \\ x &= 7\frac{1}{2}^\circ \checkmark \end{aligned}$$

$$\begin{aligned} 7. \text{ a) } \frac{1 + \sin \theta + \cos \theta}{1 + \sin \theta - \cos \theta} &= \frac{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}}{1 + \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2}} \\ &= \frac{1+t^2+2t+1-t^2}{1+t^2+2t-1+t^2} \\ &= \frac{2(t+1)}{2t(t+1)} = \frac{1}{t} \checkmark \end{aligned}$$

$$\begin{aligned} \text{b) } \frac{5(1+t^2)}{1+t^2} + \frac{6 \cdot 2t}{1+t^2} &= -5 \\ 5 - 5t^2 + 12t &= -5 - 5t^2 \\ 12t &= -10 \end{aligned}$$

$$t = -\frac{10}{12} = -\frac{5}{6} \checkmark$$

$$\tan\left(\frac{\theta}{2}\right) = -\frac{5}{6} \checkmark$$

$$\begin{aligned} \frac{\theta}{2} &\doteq 140^\circ 12' \\ \theta &= 280^\circ 23' \checkmark \end{aligned}$$

Test for 180° :

$$5x(1) + 6x(0) = -5 \checkmark$$

$\therefore$ Solutions are $180^\circ, 280^\circ 23'$

$$\begin{aligned} \textcircled{8} \sin 18^\circ &= \frac{\sqrt{5}-1}{4} \\ \cos 36^\circ &= 1 - 2\sin^2 18^\circ \checkmark \\ &= 1 - 2\left(\frac{\sqrt{5}-1}{4}\right)^2 \\ &= \frac{8 - (5 - 2\sqrt{5} + 1)}{8} \\ &= \frac{2 + 2\sqrt{5}}{8} = \frac{1 + \sqrt{5}}{4} \checkmark \end{aligned}$$

$$\begin{aligned} \textcircled{9} (x+y)(x^2+xy+y^2) &= x^3 + x^2y + xy^2 + y^3 \\ &= x^3 + y^3 \checkmark \end{aligned}$$

$$\text{b) LHS} = \cos^6 \beta + \sin^6 \beta$$

$$\begin{aligned} (\cos^2 \beta)^3 + (\sin^2 \beta)^3 &= (\cos^2 \beta + \sin^2 \beta)(\cos^4 \beta - \cos^2 \beta \sin^2 \beta + \sin^4 \beta) \\ &= 1 \times (\cos^4 \beta - 2\cos^2 \beta \sin^2 \beta + \sin^4 \beta + \sin^2 \beta \cos^2 \beta) \\ &= (\cos^2 \beta - \sin^2 \beta)^2 + \frac{1}{4} \sin^2 2\beta \checkmark \\ &= \cos^2 2\beta + \frac{1}{4} (1 - \cos^2 2\beta) \checkmark \\ &= \frac{1}{4} + \frac{3}{4} \cos^2 2\beta = \text{RHS.} \end{aligned}$$

$$10) a) \sin 3\alpha = \sin(2\alpha + \alpha)$$

$$= \sin 2\alpha \cdot \cos \alpha + \cos 2\alpha \sin \alpha \checkmark$$

$$= \sin 2\alpha \cos^2 \alpha + (1 - 2\sin^2 \alpha) \sin \alpha$$

$$\checkmark = 2\sin \alpha (1 - \sin^2 \alpha) + (1 - 2\sin^2 \alpha) \sin \alpha$$

$$= 3\sin \alpha - 4\sin^3 \alpha$$

$$b) 3(4\sin^3 \alpha - 3\sin \alpha) + 2(\cos 3\alpha + \sin 3\alpha) \checkmark$$

$$= 3(-1)\sin 3\alpha + 2(\cos 3\alpha + \sin 3\alpha)$$

$$= 2\cos 3\alpha - 2\sin 3\alpha \checkmark$$

$$= 2\sqrt{2} \left(\frac{2}{2\sqrt{2}} \cdot \cos 3\alpha - \frac{2}{2\sqrt{2}} \sin 3\alpha \right)$$

$$= 2\sqrt{2} \left(\frac{1}{\sqrt{2}} \cdot \cos 3\alpha - \frac{1}{\sqrt{2}} \sin 3\alpha \right) \checkmark$$

$$= 2\sqrt{2} \cos(3\alpha + 45^\circ) \text{ where}$$

$$= 2\sqrt{2} \cos(3\alpha + 45^\circ) \quad \alpha = \tan^{-1}(1)$$

$$c) 2\sqrt{2} \cos(3\alpha + 45^\circ) = 2$$

$$\cos(3\alpha + 45^\circ) = \frac{1}{\sqrt{2}} \checkmark$$

$$\text{acute } \angle = 45^\circ$$

$$45^\circ \leq 3\alpha + 45^\circ \leq 1125^\circ \quad \checkmark$$

$$\therefore 3\alpha + 45^\circ = 45^\circ, 315^\circ, 405^\circ, 675^\circ, 765^\circ, 1035^\circ$$

$$\alpha = 0^\circ, 90^\circ, 120^\circ, 210^\circ, 240^\circ, 330^\circ \checkmark$$